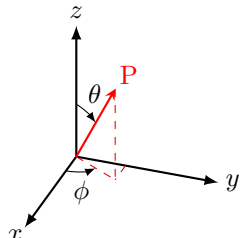
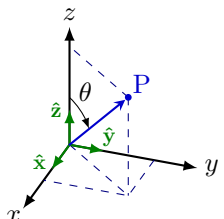
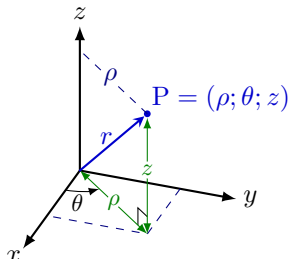
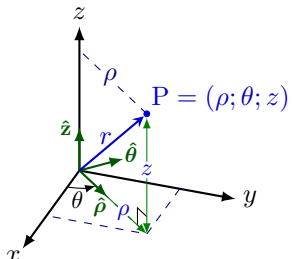
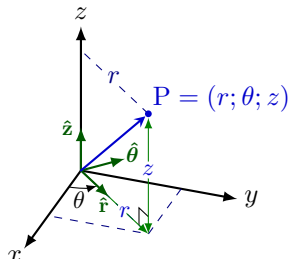
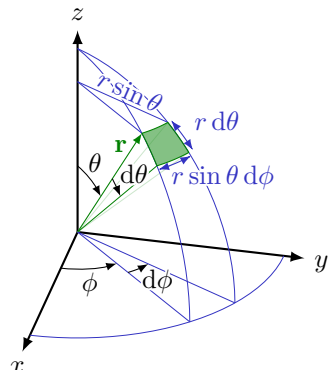

z
P
θ
y
ϕ
x

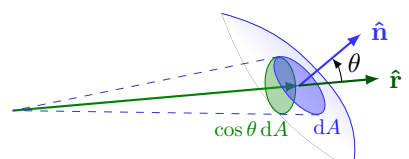

z
P
θ
ẑ
ŷ
x̂
y
x


z
ρ
P = (ρ; θ; z)
r
z
θ
ρ
y
x


z
ρ
P = (ρ; θ; z)
ẑ
r
θ̂
z
θ
ρ̂
ρ
y
x


z
r
P = (r; θ; z)
ẑ
θ̂
z
θ
r̂
r
y
x


z
r sin θ
r dθ
r
θ
dθ
r sin θ dϕ
y
ϕ
dϕ
x


n̂
θ
r̂
cos θ dA
dA